Model-Extraction Attack against FPGA-DNN Accelerator Utilizing Correlation Electromagnetic Analysis

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***Abstract*—This work presents a model-extraction attack on a DNN accelerator, which is implemented on FPGA. An adversary can get DNN model parameters by exploiting electromagnetic leakage from the accelerator during operation. Our experimental results show that the adversary can extract trained model parameters from a DNN accelerator even if the DNN model parameters are protected with data encryption. This suggests that countermeasures against side-channel leaks are important when implementing a DNN accelerator on FPGA or ASIC.**

I. Introduction

Training a deep neural network (DNN) model requires a huge dataset, computing resources, and expertise. Since trained DNN model parameters are valuable, adversaries have an incentive to steal them. In particular, when a DNN is executed on an edge device, adversaries can derive the weight stream by directly tapping the datapath of off-chip memory. Memory encryption is a good countermeasure for it. However, some papers show that an adversary can steal a DNN model information from side-channel analysis even if the model is stored with encryption [1], [2]. In this paper, we propose a model-extraction attack that steals weight parameters from Multi-layer Perceptron (MLP) models implemented on FPGA. It is the first model-extraction attack that utilizes electromagnetic (EM) leakage from DNN accelerators implemented on hardware.

II. Attacking Scenario

The goal of the adversary is to recover the MLP architecture and weight parameters. The adversary utilizes correlation electromagnetic analysis (CEMA) which is a typical and powerful method for stealing a cryptographic key using EM leakage from a cryptographic circuit under operation.

Scenario: We suppose an MLP network that is implemented on FPGA. The input to the MLP is controlled by the adversary, and model-parameters are hidden on the off-chip memory. The adversary tries to recover the model parameters by probing EM leakage under MLP processing.

Settings: We implemented a 3-layer MLP composed of two input nodes, five hidden layer nodes and two output nodes. The MLP has 20 weight parameter that the adversary tries to steal. The parameters are expressed by 8-bit fixed point. The SOP (Sum of Product) calculations of MLP are iteratively performed on multiplier and accumulator-register on processing unit as shown in Fig.1.

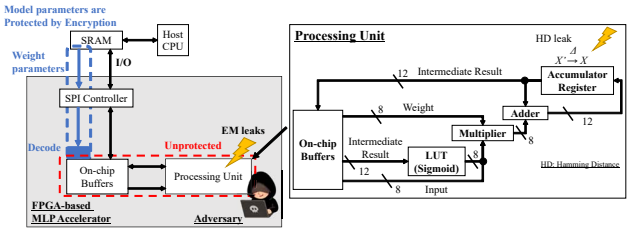


Fig. 1. A scenario of model-extraction attack

III. Experimental Result and Discussion

The adversary exploits the correlation between EM leakage and the hamming-distance of an accumulator-register, which is the number of flipped bits when the accelerator writes intermediate results on SOP calculation. The experimental results show that the adversary identified the 19 out of 20 weight parameters from 60,000 traces that measured at the single probing position. Although these results are preliminary experiments with very simple DNN accelerators, the potential of a model-extraction attack by CEMA has been demonstrated. Hence, the threat of CEMA attacks should be considered for the secure-design on future DNN accelerators.

References

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